AITS
ALL INDIA TEST SERIES
SELECTION HOGA YAHIN SE
Prayas JEE AIR 2025

Dropper JEE Main

AJPM6/05

Weekly Test 30
(AITS 05)

DURATION : 180 Minutes

DATE : 12/01/2025

M.MARKS : 300

ANSWER KEY

PHYSICS

1. (4)
2. (1)
3. (2)
4. (3)
5. (3)
6. (1)
7. (1)
8. (2)
9. (3)
10. (2)
11. (4)
12. (3)
13. (4)
14. (1)
15. (2)
16. (1)
17. (1)
18. (3)
19. (2)
20. (2)
21. (5)
22. (3)
23. (18)
24. (25)
25. (96)

CHEMISTRY

26. (3)
27. (2)
28. (2)
29. (3)
30. (1)
31. (3)
32. (1)
33. (2)
34. (2)
35. (3)
36. (4)
37. (1)
38. (1)
39. (4)
40. (2)
41. (3)
42. (3)
43. (2)
44. (4)
45. (3)
46. (7)
47. (11)
48. (7)
49. (10)
50. (4)

MATHEMATICS

51. (1)
52. (1)
53. (4)
54. (3)
55. (3)
56. (2)
57. (4)
58. (2)
59. (4)
60. (2)
61. (1)
62. (2)
63. (4)
64. (2)
65. (3)
66. (4)
67. (2)
68. (1)
69. (4)
70. (2)
71. (13)
72. (3)
73. (65)
74. (20)
75. (386)

SECTION-I (PHYSICS)

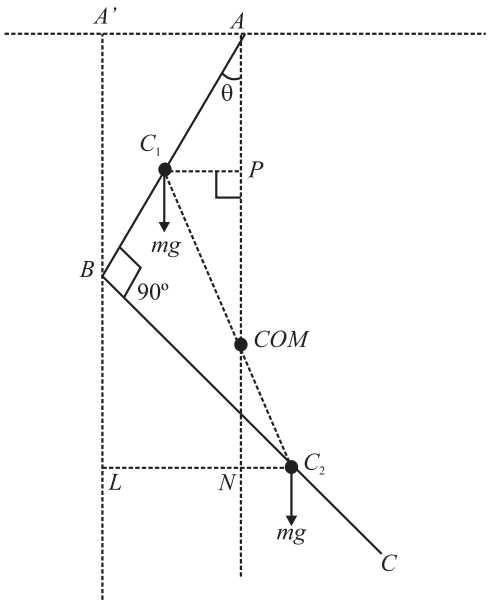
1. **(4)**

Lets consider mass of each rod is m.

For stable equilibrium the torque about point *O* should be zero

$$mg\frac{a}{2}\sin\theta = mg\left(a\cos\theta - a\sin\theta\right)$$

$$\tan\theta = \frac{2}{3}$$



$$\Rightarrow \quad \tan^{-1}\left(\frac{2}{3}\right)$$

[9 Jan, 2019 (Shift-I)]

2. **(1)**

$ME = PE + KE = 10\text{J}$

At x_3, $10 = KE + 4 \Rightarrow KE = 6\text{J}$

At x_2, $10 = KE + 0 \Rightarrow KE_{max} = 10\text{J}$

At $x < x_1$, $10 = KE + 8$ KE is 2J

[27 July, 2021 (Shift-II)]

3. **(2)**

Applying conservation of linear momentum let v' be the final speed of the system

$Mv = (2m + M)v'$

$$v' = \frac{mv}{2m+M}$$

Initial kinetic energy = $\frac{1}{2}mv^2$

Final kinetic energy

$$= \frac{1}{2}(2m+M)\left(\frac{mv}{2m+M}\right)^2$$

Initial kinetic energy – Final kinetic energy = $\frac{3}{4}$ of initial kinetic energy

$$\frac{1}{2}mv^2 - \frac{1}{2}(2m+M)\left(\frac{mv}{2m+M}\right)^2 = \frac{3}{4}\times\frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 - \frac{3}{8}mv^2 = \frac{1}{2}(2m+M)\frac{m^2v^2}{(2m+M)^2}$$

$$\frac{mv^2}{8} = \frac{m^2v^2}{2(2m+M)}$$

$$\Rightarrow \quad 4m = 2m + M$$

$$\Rightarrow \quad \frac{m}{M} = \frac{1}{2}$$

[9 Jan, 2019 (Shift-I)]

4. **(3)**

Net force on particle towards centre of circle is

$$F_C = \frac{GM^2}{2(2a)^2} + \frac{GM^2}{(2a)^2}\sqrt{2}$$

$$= \frac{GM^2}{(2a)^2}\left(\frac{1}{2} + \sqrt{2}\right)$$

This force will act as centripetal force. Distance of particle from centre of circle is $\frac{2a}{\sqrt{2}}$

$$r = \frac{2a}{\sqrt{2}}, F_c = \frac{Mv^2}{r}$$

$$\frac{Mv^2}{2a/\sqrt{2}} = \frac{GM^2}{(2a)^2}\left(\frac{1}{2} + \sqrt{2}\right)$$

$$v^2 = \frac{GM}{2a}\left(\frac{1}{2\sqrt{2}} + 1\right)$$

$$v^2 = \frac{GM}{2a}(1.35)$$

$$v = 1.16\sqrt{\frac{GM}{2a}} = 0.82\sqrt{\frac{GM}{a}}$$

[8 April, 2019 (Shift-I)]

5. **(3)**

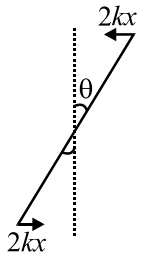
Heat lost = Heat gain

$$\Rightarrow \quad M_2 \times 1 \times 50 = M_1 \times 0.5 \times 5 + M_1 L$$

$$\Rightarrow \quad L = \frac{50M_2 - 2.5M_1}{M_1} = \frac{50M_2}{M_1} - 2.5$$

[12 April, 2019 (Shift-I)]

6. **(1)**



Torque on rod at displacement θ from mean position θ is very small

$$x = \frac{L}{2}\theta$$

$$\tau = 2(2k)x\frac{L}{2} = 2(2k)\frac{L_2}{4}\theta = \frac{(2k)L^2}{2}\theta$$

Now, $\tau = i\alpha$

$$\frac{(2k)L^2}{2}\theta = \frac{mL^2}{12}\alpha;\ \alpha = \frac{6(2k)}{m}\theta$$

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{12k}{m}} = \frac{1}{\pi}\sqrt{\frac{3k}{m}}$$

[12 Jan, 2019 (Shift-I)]

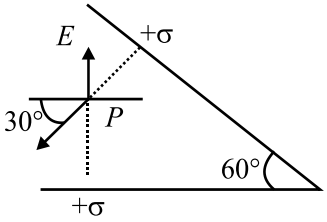
7. (1)

Electric field due to each sheet $\vec{E} = \frac{\sigma}{2\varepsilon_0}$

So, net electric field between plates

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\cos 30^\circ(-\hat{x})\left[\frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}\sin 30^\circ\right](\hat{y})$$

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\left[\left(-\frac{\sqrt{3}}{2}\right)\hat{x} + \frac{1}{2}\hat{y}\right]$$



[7 Jan, 2020 (Shift-I)]

8. (2)

$$\frac{1}{2}mv^2 = mgy + \Delta U_{elec}$$

$$\Delta U_{elec} = kQq\left[\frac{1}{R} - \frac{1}{(R+y)}\right]$$

$$\Rightarrow \quad v^2 = 2y\left[\frac{qQ}{4\pi\varepsilon_0 R(R+y)m} + g\right]$$

[5 Sep, 2020 (Shift-I)]

9. (3)

Power in external $R = \left(\frac{\varepsilon}{r+R}\right)^2 R$

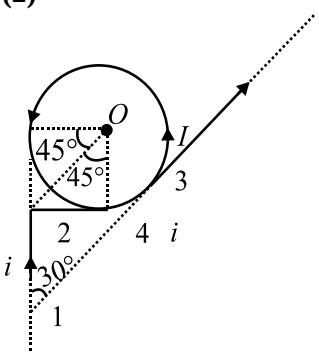
This power is maximum, when R = r

$$r = \int_a^b \rho\frac{dx}{2\pi xl}$$

$$r = \frac{\rho}{2\pi l}\ln(3)$$

[3 Sep, 2020 (Shift-I)]

10. (2)



The net magnetic field

$$= \frac{\mu_0 i}{4\pi R}[\sin 90^\circ - \sin 45^\circ]\otimes + \frac{\mu_0 i}{4\pi R}(\sin 45^\circ + \sin 0^\circ)\odot$$

$$+ \frac{\mu_0 i}{4\pi R}\left(\frac{7\pi}{3}\right)\odot + \frac{\mu_0}{4\pi}\frac{i}{R}\odot$$

$$= \frac{\mu_0 i}{4\pi R}\left[\sqrt{2} + \frac{7\pi}{3}\right]\odot$$

[8 Jan, 2020 (Shift-II)]

11. (4)

Current sensitivity = voltage sensitivity ×R

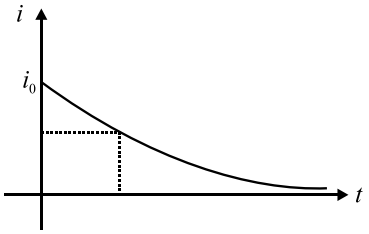
Since current sensitivity is made 1.75 times. Therefore, R also increases 1.75 time

$$= \frac{1.75 \text{ current sensitivity}}{1.75\text{R}} = \text{no change}$$

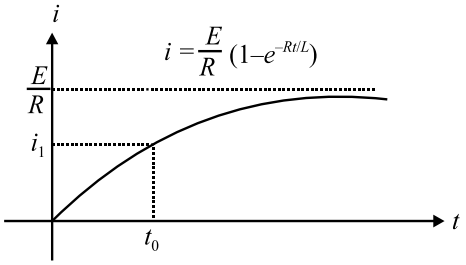
[31 Jan, 2023 (Shift-II)]

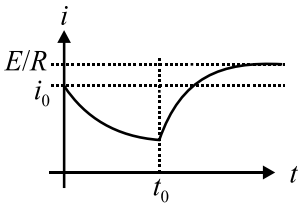
12. (3)

When S_2 is closed, current in circuit decay as



When at t_0, S_2 is opened & S_1 is closed then current grows as





[11 Jan, 2019 (Shift-I)]

13. (4)

$T_2 = 2T_1$

$u_2 = 2u_1$

$H \propto u^2$

$$\Rightarrow \quad \frac{H_1}{H_2} = \frac{1}{4}$$

14. (1)

Given $E = 30\cos(2\times10^{10}t - 100x)V/m$

$\Rightarrow \quad \omega = 2 \times 10^{11}$, K = 100

Wave speed = $\frac{2\times10^{10}}{100} = 2\times10^{8}$ m/s

Refractive Index = $\frac{C}{\text{speed}} = \frac{3\times10^{8}}{2\times10^{8}} = \frac{3}{2}$

$\frac{3}{2} = \sqrt{\varepsilon_r \times 1}$

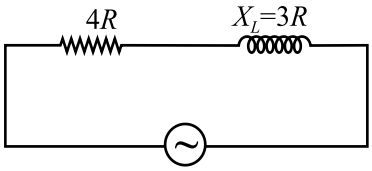
[$\therefore$ As refractive index = $\sqrt{\mu_r \varepsilon_r}$]

$\varepsilon_r = \frac{9}{4} = 2.25$

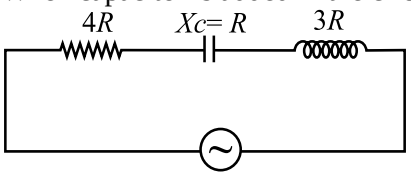
[1 Sep, 2021 (Shift-II)]

15. (2)

Initially, $\cos\phi_1 = \frac{4R}{\sqrt{(4R)^2 + (3R)^2}} = \frac{4R}{5R} = \frac{4}{5}$

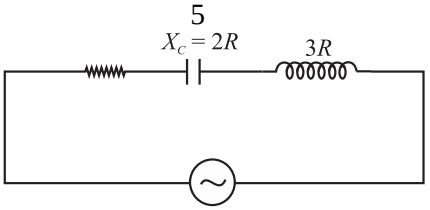


When capacitor is added in the circuit



$\cos\phi_2 = \frac{4R}{\sqrt{(4R)^2 + (3R-R)^2}} = \frac{4R}{\sqrt{20}R}$

$\Rightarrow \quad \frac{\cos\phi_2}{\cos\phi_1} = \frac{\frac{4}{\sqrt{20}}}{\frac{4}{5}} = \sqrt{\frac{25}{20}} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$



Hence, the correct value of x in the ratio $\sqrt{5} : x$ is 2

[27 Aug, 2021 (Shift-II)]

16. (1)

$R = \frac{mv}{qB} = \frac{\sqrt{2m(K.E.)}}{qB}$

$R = \frac{\sqrt{2\times9.1\times10^{-31}\times(2.88\times10^{3}\times1.6\times10^{-19})}}{1.6\times10^{-19}\times3.01\times10^{-3}}$

R = 6 cm

$\sin\theta = \frac{3}{6} = 0.5$

$\tan\theta = \frac{QU}{TU}$

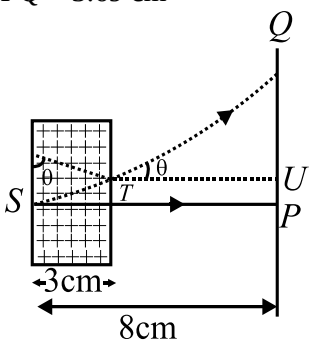
$\frac{1}{\sqrt{3}} = \frac{QU}{5}$

$QU = \frac{5}{\sqrt{3}}$

$PQ = R(1 - \cos\theta) + \frac{5}{\sqrt{3}}$

$PQ = 6\left(1 - \frac{\sqrt{3}}{2}\right) + \frac{5}{\sqrt{3}}$

PQ = 3.69 cm



[12 April, 2019 (Shift-II)]

17. (1)

We know that

$\frac{hc}{\lambda} = \phi + eV_s$

$\therefore \quad \frac{1240}{500} = \phi + 0.62$ ……(i)

$\frac{1240}{\lambda} = \phi + 1.86$ ……(ii)

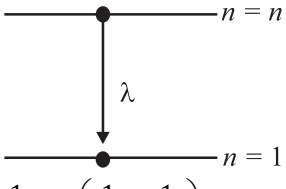
Using above two equation we can calculate:

λ = 333 nm

[25 Feb, 2021 (Shift-II)]

18. (3)

According to definition of wave number, we can write



$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$

$\Rightarrow \quad 1 - \frac{1}{n^2} = \frac{1}{\lambda R}$

$\Rightarrow \quad \frac{1}{n^2} = 1 - \frac{1}{\lambda R} = \frac{\lambda R - 1}{\lambda R}$

$\Rightarrow \quad n = \sqrt{\frac{\lambda R}{\lambda R - 1}}$

[25 July, 2022 (Shift-II)]

19. (2)

Density of Nucleus $= \frac{\text{Mass}}{\text{Volume}}$

$= \frac{1.67\times10^{-27}\times A}{\frac{4}{3}\pi(1.3)^3\times10^{-45}\times A} = 1.81\times10^{17}\ \text{kg/m}^3$

[3 Sep, 2020 (Shift-II)]

20. (2)

In – *ve* half cycle

$D_1 \rightarrow R.B.$; $D_2 \rightarrow F.B.$

$0 - 0.6\ V$

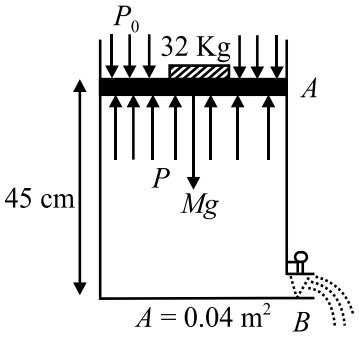
V_{out} same as Vin

In + *ve* half cycle

$D_2 \rightarrow R.B.$; $D_1 \rightarrow F.B.$

[28 June, 2022 (Shift-II)]

21. (5)



Force at point *A*,

$mg + p_0A = pA \Rightarrow p = p_0 + \frac{mg}{A}$

Given, area of $A = 0.4\ m^2 = 0.4 \times 10^4\ cm^2$ and area of $B = 1\ cm^2$

Applying continuity equation

$Av_1 = av$ [V_1 = Velocity at point *A*]

$\Rightarrow \quad v_1 = \frac{a}{A}v$

As $A >> a$ so $\frac{a}{A}$ very small

$\therefore \quad v_1 << ev$

So we can neglect v_1 by assuming $v_1 = 0$

On applying Bernoulli's theorem at points *A* and *B*,

$$p_0 + \frac{mg}{A} + \frac{\rho v_1^2}{2} + \rho gh = p_0 + \frac{\rho v^2}{2} + 0$$

$$\Rightarrow \quad p_0 + \frac{mg}{A} + 0 + \rho gh = p_0 + \frac{\rho v^2}{2}$$

$$\Rightarrow \quad \frac{mg}{A} + \rho gh = \frac{\rho v^2}{2}$$

$v = 5$ m/s

[18 March, 2021 (Shift-II)]

22. (3)

$$I_1 = I_3 = \frac{Ma^2}{4 \times 4}, I_2 = I_4 = \frac{Ma^2}{4 \times 4} + \frac{Ma^2}{4}$$

$$I_{00'} = I_1 + I_2 + I_3 + I_4$$

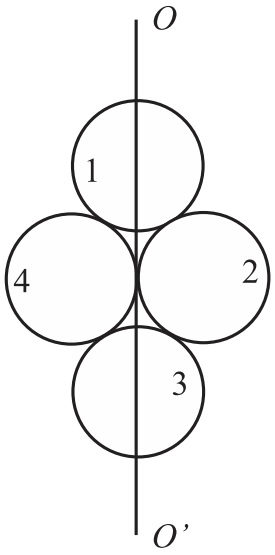
$$I_{00'} = \frac{6Ma^2}{8} = \frac{3}{4}Ma^2$$

$$I_{00'} = 4Mk^2$$

$$\frac{3}{4}Ma^2 = (4M)k^2$$

$$k^2 = \frac{3}{16}a^2$$

$$k = \frac{\sqrt{3}a}{4}$$



$\Rightarrow \quad x = 3$

[28 July, 2022 (Shift-I)]

23. (18)

$$C = \frac{A\varepsilon_0}{d - t + \frac{t}{k}}$$

$$\Rightarrow \quad C = \frac{7 \times 8.85 \times 10^{-12}}{3.5 \times 10^{-3}} = 17.7\,\text{nF}$$

[18 March, 2021 (Shift-I)]

24. (25)

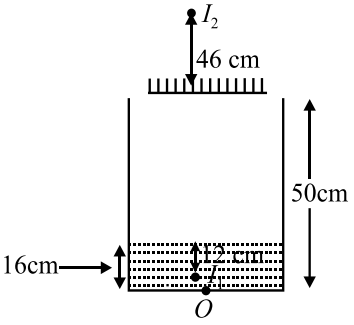
Force needed to move the rod is $F = i \ell B$

$$= \left(\frac{vBl}{R}\right)Bl = \frac{vB^2l^2}{R} = \frac{8}{5} \times \frac{625}{10^4} \times \frac{1}{4}$$

$$F = 25 \times 10^{-3}\,\text{N}$$

[10 April, 2023 (Shift-I)]

25. (96)



Apparent depth of $O = \frac{\text{real depth}}{\text{refractire index}} = \frac{d}{\mu} = 12$

Distance between *O* and $I_2 = 46 + 50 = 96\ cm$

[11 April, 2023 (Shift-II)]

SECTION-II (CHEMISTRY)

26. (3)

30% (w/w) aqueous solution means,

100g solution contains 30 g KI

$\therefore$ Mass of solvent = 100 – 30 = 70g

$$\text{Moles of KI} = \frac{30}{166} = 0.18$$

$$\text{Moles of } H_2O = \frac{70}{18} = 3.89$$

$$x_{KI} = \frac{0.18}{0.18 + 3.89} = 0.044$$

[9 April, 2019 (Shift-II)]

27. (2)

$$r = \frac{n^2}{Z} \times a_0 = \frac{5^2}{4} \times a_0 = \frac{25}{4} a_0$$

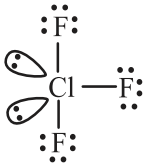
[8 Jan, 2020 (Shift-II)]

28. (2)

E.C. of I is $1s^2\ 2s^2\ 2p^3$, which is half-filled but EC of II is $1s^2\ 2s^2\ 2p^4$, which is more than half filled so, IP_I of I > I.P_I of II

[26 Feb, 2021 (Shift-I)]

29. (3)



Shape $\rightarrow$ T shape

Geometry $\rightarrow$ TBP

[29 June, 2022 (Shift-II)]

30. (1)

(i) $H_2(g) + I_2(g) \rightleftharpoons 2HI\ (g);\ \Delta n_g = 0$

$$K_p = K_c(RT)^\circ \Rightarrow \frac{K_p}{K_c} = 1$$

(ii) $2NO\ (g) + O_2(g) \rightleftharpoons 2NO_2(g);$

$\Delta n_g = 2 - (2 + 1) = -1$

$$\therefore \quad \frac{K_p}{K_c} = (RT)^{-1} = \frac{1}{RT} = \frac{1}{24.62}$$

$$= 0.0406\ L^{-1}atm^{-1}mol$$

(iii) $PCl_5(g) \rightleftharpoons PCl_3\ (g) + Cl_2(g)$

$\Delta n_g = 2 - 1 = 1;$

$$\frac{K_p}{K_c} = RT = 24.62\ L\ atm.mol^{-1}$$

[10 Jan, 2019 (Shift-I)]

31. (3)

O.N. of Cr in CrO_5 is + 6, as it has 2–peroxy bonds and in $Cr(CO)_6$ is O.

[17 March, 2021 (Shift-II)]

32. (1)

It is the electronic configuration of $_5B$. Its diagonal element is $_{14}Si$; whose E.C. is $[Ne]3s^23p^2$

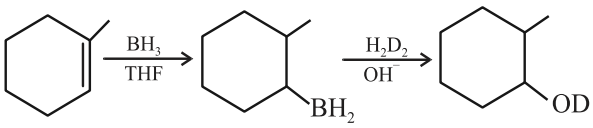
[20 July, 2021 (Shift-II)]

33. (2)

2–Bromo–5–oxohexanoic acid

[8 April, 2023 (Shift-II)]

34. (2)



[12 April, 2019 (Shift-I)]

35. (3)

Polarity: Ethanol > n-Butanol > Ethoxy ethane: Non-polar solvent is in higher composition So non-polar compound will separate out first.

[7 Jan, 2020 (Shift-II)]

36. (4)

$$\left(\frac{\Delta E}{\Delta T}\right) = \frac{\Delta S}{nF}$$

A cell with minimal variation in EMF can be used across a broad temperature range with minimal deviation from the standard value.

[26 June, 2022 (Shift-I)]

37. (1)

0.20 = K (10) (0.25) .....(i)

0.80 = K (a) (0.25) .....(ii)

1.60 = K (5) (b) ……(iii)

$$\frac{r_2}{r_1} = \frac{0.80}{0.20} = \frac{a}{10}$$

$$\frac{r_3}{r_2} = \frac{1.60}{0.8} = \frac{5 \times b}{40 \times 0.25} = \frac{b}{2}$$

$\therefore$ a = 40

$\therefore$ b = 4

[11 April, 2023 (Shift-II)]

38. (1)

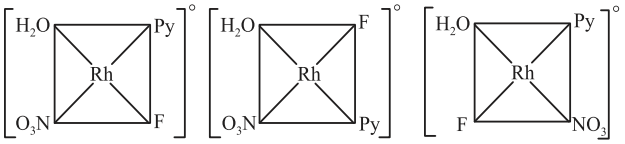
$$2\overset{+5}{Pb}(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4\overset{+4}{N}O_2 + O_2$$

$$\overset{+5}{N}O_3^- \longrightarrow \overset{+4}{N}O_2 \quad \Delta n = +1 \text{ Only}$$

[4 Sep, 2020 (Shift-I)]

39. (4)

3, Isomers



[10 Jan, 2019 (Shift-I)]

40. (2)

NO is paramagnetic & Neutral oxide

[17 March, 2021 (Shift-II)]

41. (3)

NaSH, Nu^- which favours S_N2 reaction and S_N2 reactions are faster with better leaving group.

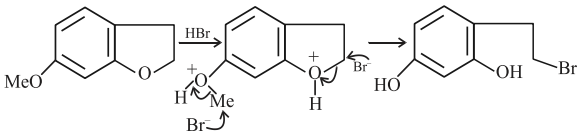
[29 July, 2022 (Shift-I)]

42. (3)

Phenol is a very weak acid so it does not react with weak base $NaHCO_3$ while 2,4-Dimitri phenol is comparatively stronger due to the-M effect of NO_2 and produces CO_2 when reacts with $NaHCO_3$

[31 Jan, 2024 (Shift-I)]

43. (2)



[2 Sep, 2020 (Shift-I)]

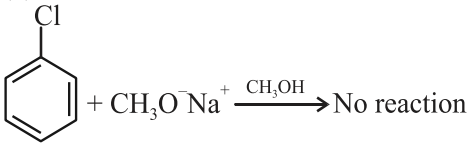
44. (4)

	Reaction		Reagents
A	Hoffmann degradation	$\rightarrow$	Br_2, NaOH
B	Clemenson reduction	$\rightarrow$	Zn–Hg/HCl
C	Cannizaro-reaction	$\rightarrow$	Conc. KOH, Δ
D	Reimer-Tiemann reaction	$\rightarrow$	$CHCl_3$, NaOH/ H_3O^+

$\therefore$ Correct match is (A) –R, (B)– S, (C) – P, (D) – Q

[29 Jan 2023 (Shift-I)]

45. (3)



$\because$ Nucleophilic aromatic substitution reaction doesn't takes place in Williamson ether synthesis, instead it proceeds through S_N2 mechanism.

[27 Aug, 2021 (Shift-I)]

46. (7)

For an adiabatic process, q = 0

$\Delta U = w$, n = 2

w = 2 kJ = 2000 J

$\Delta U = nC_v\Delta T$

$-2000 = 2 \times 50 (T_2 - 300)$

$T_2 = 280$ K

= 7°C

[30 Jan, 2023 (Shift-II)]

47. (11)

$$M(H_2SO_4) = \frac{49}{98 \times 0.1} = 5M$$

$$M(NaOH) = \frac{20}{40 \times 0.1} = 5\ M$$

80 L of X = $80 \times 5 \times 10^{-3} = 4 \times 10^{-1}$ mol of NaOH

20 L of Y = $20 \times 5 \times 10^{-3} = 10^{-1}$ mol of H_2SO_4

$2\ NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

t = 0,	4×10^{-1}	10^{-1}	0	0
t = t	2×10^{-1}	0	10^{-1}	2×10^{-1}

Total volume = 100 L

Moles of OH^- = 2×10^{-1}

$$[OH^-] = \frac{2\times10^{-1}}{100} = 2\times10^{-3}$$

pOH = 2.7

pH = 14 – 2.7

$\approx$ 11

[7 Jan, 2020 (Shift-I)]

48. (7)

$$E = E^0 - \frac{0.0591}{2}\log\frac{1}{[H^+]^2}$$

$$= 1.23 + \frac{0.0591}{2}\log[H^+]^2$$

= 1.23 – 0.0591 × pH

= 1.23 – 0.0591 × 9

= 0.6981 $\approx$ 0.7

$\therefore$ Value 7×10^{-1}

x = 7

[8 Jan, 2020 (Shift-I)]

49. **(10)**

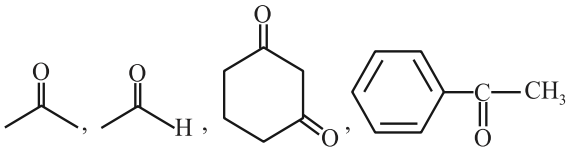
Compound 'A' $\Rightarrow$ phenol

'Z' $\Rightarrow$ 2, 4, 6 – Tribromophenol

Total no. of C + O + Br = 6 + 1 + 3 = 10

[26 June, 2022 (Shift-I)]

50. **(4)**



[1 Feb, 2023 (Shift-I)]

SECTION-III (MATHEMATICS)

51. **(1)**

$a + b + c + d = 11$

$$\Rightarrow \frac{a}{5}+\frac{a}{5}+\frac{a}{5}+\frac{a}{5}+\frac{a}{5}+\frac{b}{3}+\frac{b}{3}+\frac{b}{3}+\frac{c}{2}+\frac{c}{2}+d=11$$

$$\frac{5\left(\frac{a}{5}\right)+3\left(\frac{b}{3}\right)+2\left(\frac{c}{2}\right)+d}{11} \geq \left(\frac{a^5b^3c^2d}{5^5 3^3 2^2}\right)^{1/11}$$

$$1 \geq \left(\frac{a^5b^3c^2d}{5^5 3^3 2^2}\right)^{1/11}$$

$a^5b^3c^2d^2 \leq 337500$. Hence, $\beta = 54$

[11 April, 2023 (Shift-II)]

52. **(1)**

For $x \leq 3$ or $x \geq 5$

$x^2 - 8x + 15 - 2x + 7 = 0$

$x = 5 + \sqrt{3}$ or $x = 5 - \sqrt{3}$ (reject)

For $3 < x < 5$, $x^2 - 8x + 15 + 2x - 7 = 0$

$x = 4$ or $x = 2$ (reject)

Hence, sum of all the roots

$$= \left(5+\sqrt{3}\right)^2 + 4^2 = \left(44 + 10\sqrt{3}\right)$$

[6 April, 2023 (Shift-I)]

53. **(4)**

$x^2 - 3x + [x] + 2 = x[x]$

$\Rightarrow x^2 - 3x + 2 = x[x] - [x]$

$\Rightarrow (x - 1)(x - 2) = [x] \,.\, (x - 1)$

$\Rightarrow x = 1$ or $x - 2 = [x]$

$\Rightarrow x - [x] = 2$

$\Rightarrow \{x\} = 2$ (Not Possible) ({.} fractional part of x)

Only one solution $x = 1$ in $(-\infty, \infty)$

[24 Jan, 2023 (Shift-I)]

54. **(3)**

$$\vec{\lambda} = -\hat{i} + 2\hat{j} + \lambda\left(\hat{i} + 3\hat{k}\right)$$

$\therefore$ drs of required line are < 1, 0, 3 >

Hence $\frac{x}{1} = \frac{y+2}{0} = \frac{z-5}{3}$

[10 Jan, 2019 (Shift-II)]

55. **(3)**

Let equation of circle

$x^2 + y^2 - hx - hy = 0$ ….(1)

given circle

$x^2 + y^2 - 4x - 6y + 10 = 0$ …..(2)

both are orthogonal

$2g_1g_2 + 2f_1f_2 = c_1 + c_2$

apply and find 'h'

56. **(2)**

$z = \alpha + i\beta$

Given, $|z + 2| = z + 5(1 + i)$

$$\Rightarrow \sqrt{(\alpha+2)^2+\beta^2} = (\alpha+5) + i(\beta+5)$$

$$\Rightarrow \sqrt{(\alpha+2)^2+\beta^2} = \alpha+5 \quad \text{….(i)}$$

$\beta + 5 = 0 \Rightarrow \beta = -5$ ...(ii)

From eq. (i) $\Rightarrow \sqrt{(\alpha+2)^2+25} = \alpha+5$

$\Rightarrow \alpha^2 + 4\alpha + 29 = \alpha^2 + 10\alpha + 25 \Rightarrow 6\alpha = 4$

$$\Rightarrow \alpha = \frac{2}{3}$$

$3\alpha + \beta = -3$, $3\alpha\beta = -10$

Equation with roots – 3 and – 10 is

$x^2 + 13x + 30 = 0$

[8 April, 2023 (Shift-I)]

57. **(4)**

In VANCOUVER word, total letter = 9

V comes 2 times, vowels (A, E, O, U) do not come together

$$= \frac{9!}{2!} - \frac{6!4!}{2!}$$

$$= \frac{6!}{2}\left(9 \cdot 8 \cdot 7 - 24\right)$$

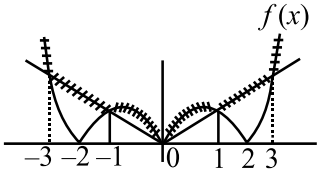
= 240(6!)

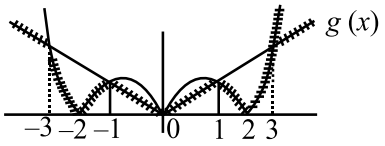
[8 April, 2023 (Shift-II)]

58. **(2)**

$f(x)$ is non-differentiable at $x = -3, -1, 0, 1, 3$

and $g(x)$ is non-differentiable at $x = -3, -2, -1, 0, 1, 2, 3 \Rightarrow$ (2)





59. **(4)**

$A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$R = \{(1, 6), (6, 1) (2, 5) (5, 2) (3, 4) (4, 3)\}$

Reflexive: $aRa \rightarrow$ not possible

$\therefore$ (1, 1) not in R

Symmetric: $aRb \Leftrightarrow bRa$

True

Transitive: $aRb, bRc \Leftrightarrow aRc$

Hence, (1, 6), (6, 1) But (1, 1) $\notin R$

Hence, it is not transitive.

[8 April, 2023 (Shift-II)]

60. **(2)**

$P = x(66 - x)$, Quadratic expression's maximum value exists at $x = 33$

$M = 33 \times 33$

$$x(75-x) \geq \frac{6}{11} \times 33 \times 33$$

$9 \leq x \leq 66$

A : {9, 12, 15, 18, ... 66}

$$P(A) = \frac{20}{58} = \frac{10}{29}$$

[25 Jan, 2023 (Shift-I)]

61. **(1)**

For $x^2 + 2\alpha x + \beta > 0 \ \forall \ x \in R$ to hold, we should have $4\alpha^2 - 4\beta < 0$

If $\alpha = 1$, β can be 2, 3, 4, 5, 6 i.e., 5 choices

If $\alpha = 2$, β can be 5, 6 i.e., 2 choices

If $\alpha = 3, 4, 5, 6$, No possible value for β i.e., 0 choices

Hence total favorable outcomes

$= 5 + 2 + 0 + 0 + 0 + 0 = 7$

Total possible choices for α and $\beta = 6 \times 6 = 36$

$$\text{Required probability} = \frac{7}{36}$$

[25 Jul, 2022 (Shift-I)]

62. **(2)**

Given data

$(1 - x^2y^2)\, dx = ydx + xdy$, $y(1) = 2$

$y(2) = \alpha$

$$dx = \frac{d(xy)}{1-(xy)^2} \Rightarrow \int dx = \int \frac{d(xy)}{1-(xy)^2}$$

$$x = \frac{1}{2}\ln\left|\frac{1+xy}{1-xy}\right| + c$$

Put $x = 1$ and $y = 4$

$$1 = \frac{1}{2}\ln\left|\frac{1+4}{1-4}\right| + c$$

$$\Rightarrow \quad c = 1 - \frac{1}{2}\ln\left(\frac{5}{3}\right)$$

So, $x = \frac{1}{2}\ln\left(\frac{1+xy}{1-xy}\right) + 1 - \frac{1}{2}\ln\left(\frac{5}{3}\right)$ ....(i)

Now put $x = 2$, $y = \alpha$ in equation (i)

$$2 = \frac{1}{2}\ln\left|\frac{1+2\alpha}{1-2\alpha}\right| + 1 - \frac{1}{2}\ln\left(\frac{5}{3}\right)$$

$$\Rightarrow \quad 1 + \frac{1}{2}\ln\left(\frac{5}{3}\right) = \frac{1}{2}\ln\left|\frac{1+2\alpha}{1-2\alpha}\right|$$

$$\Rightarrow \quad 2 + \ln\frac{5}{3} = \ln\left(\frac{1+2\alpha}{1-2\alpha}\right)$$

$$\Rightarrow \quad \left|\frac{1+2\alpha}{1-2\alpha}\right| = \left(\frac{5}{3}\right)e^2$$

$$\Rightarrow \quad \frac{1+2\alpha}{1-2\alpha} = \left(\frac{5}{3}\right)e^2, -\left(\frac{5}{3}\right)e^2$$

$$\Rightarrow \quad \frac{1+2\alpha}{1-2\alpha} = \left(\frac{5}{3}\right)e^2 \Rightarrow \alpha = \frac{5e^2-3}{2\left(5e^2+3\right)}$$

And $\frac{1+2\alpha}{1-2\alpha} = -\frac{5}{3}e^2 \Rightarrow \alpha = \frac{5e^2+3}{2\left(5e^2-3\right)}$

[11 April, 2023 (Shift-I)]

63. **(4)**

Given, $P^T = P$, $Q^T = -Q$

$$A = \frac{\left(A + A^T\right) + \left(A - A^T\right)}{2}$$

$$\Rightarrow \quad P = \frac{A + A^T}{2} \text{ and } Q = \frac{A - A^T}{2}$$

$$\text{Now, P} = \frac{1}{2}\left(\begin{bmatrix} 2 & 3 \\ \alpha & 0 \end{bmatrix} + \begin{bmatrix} 2 & \alpha \\ 3 & 0 \end{bmatrix}\right)$$

$$= \frac{1}{2}\begin{bmatrix} 4 & 3+\alpha \\ \alpha+3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & \frac{\alpha+3}{2} \\ \frac{\alpha+3}{2} & 0 \end{bmatrix}$$

$$Q = \frac{A - A^T}{2} = \frac{1}{2}\begin{bmatrix} 0 & 3-\alpha \\ \alpha-3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{3-\alpha}{2} \\ \frac{\alpha-3}{2} & 0 \end{bmatrix}$$

$$\det(Q) = 36 \Rightarrow \frac{(\alpha-3)^2}{4} = 36 \Rightarrow \alpha - 3 = \pm 12$$

$\Rightarrow \alpha = 15, -9$

$\therefore$ For $\alpha = 15$

$$\det(P) = \begin{vmatrix} 2 & 9 \\ 9 & 0 \end{vmatrix} = -81$$

and For $\alpha = -9$

$$\det(P) = \begin{vmatrix} 2 & -3 \\ -3 & 0 \end{vmatrix} = -9$$

$\therefore$ The modulus of sum of P is 90.

[20 July, 2021 (Shift-I)]

64. (2)

Given that $\cos x + \cos y - \cos(x+y) = \frac{3}{2}$

$\Rightarrow 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) - \left(2\cos^2\left(\frac{x+y}{2}\right)-1\right)$

$= \frac{3}{2}$

$\Rightarrow 2\cos\left(\frac{x+y}{2}\right)\left\{\cos\left(\frac{x-y}{2}\right) - \cos\left(\frac{x+y}{2}\right)\right\} = \frac{3}{2} - 1$

$\Rightarrow \quad 2\cos\left(\frac{x+y}{2}\right)\cdot 2\sin\left(\frac{x}{2}\right)\sin\left(\frac{y}{2}\right) = \frac{1}{2}$

$\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x}{2}\right)\sin\left(\frac{y}{2}\right) = \frac{1}{8}$

As $0 < x, y < \pi$, possibility is $\frac{x}{2} = 30°$; $\frac{y}{2} = 30°$

$\Rightarrow \quad x = 60°; y = 60°$

Then, $\sin x + \cos y - \tan y$

$= \sin 60° + \cos 60° - \tan 60° = \frac{1+\sqrt{3}}{2} - \sqrt{3}$

$= \frac{1-\sqrt{3}}{2}$

[25 Feb, 2021 (Shift-II)]

65. (3)

Maximum value of $f(u)$ can be 6

$1 \le f(1) + f(2) - 3 = f(4) \le 6$

$4 \le f(1) + f(2) \le 9$

Case-(*i*) $f(1) = 1 \Rightarrow f(2) = 3, 4, 5, 6$

$\Rightarrow$ 4 mappings

Case-(*ii*) $f(1) = 2 \Rightarrow f(2) = 2, 3, 4, 5, 6$

$\Rightarrow$ 5 mappings

Case-(*iii*) $f(1) = 3 \Rightarrow f(2) = 1, 2, 3, 4, 5, 6$

$\Rightarrow$ 6 mappings

Case-(*iv*) $f(1) = 4 \Rightarrow f(2) = 1, 2, 3, 4, 5$

$\Rightarrow$ 5 mappings

Case-(*v*) $f(1) = 5 \Rightarrow f(2) = 1, 2, 3, 4$

$\Rightarrow$ 4 mappings

Case-(*vi*) $f(1) = 6 \Rightarrow f(2) = 1, 2, 3$

$\Rightarrow$ 3 mappings

$f(3)$ & $f(5)$ both have 6 mappings each

Number of functions = $(4 + 5 + 6 + 5 + 4 + 3) \times 6 \times 6 = 27 \times 36 = 972$

[11 April, 2023 (Shift-II)]

66. (4)

$\lim\limits_{x\to 0}\left(\frac{1+\tan x}{1-\tan x}\right)^{2/x} \Rightarrow \left(\lim\limits_{x\to 0} e^{\left(\frac{1+\tan x-(1-\tan x)}{1-\tan x}\right)\frac{1}{x}}\right)^2$

$\Rightarrow \quad \left(\lim\limits_{x\to 0} e^{\left(\frac{2\tan x}{1-\tan x}\right)\frac{1}{x}}\right)^2 = e^4$

[2 Sep, 2020 (Shift II)]

67. (2)

$\int\limits_{\cos x}^{1} t^2 f(t)\,dt = \sin^3 x + \cos x - 1$

Differentiating both sides we get,

$-\cos^2 x\, f(\cos x)\cdot(-\sin x) = 3\sin^2 x.\cos x - \sin x$

$\Rightarrow \quad f(\cos x) = 3\tan x - \sec^2 x$

$\Rightarrow \quad f'(\cos x)(-\sin x) = 3\sec^2 x - 2\sec^2 x \tan x$

When $\cos x = \frac{1}{\sqrt{5}}; \sin x = \frac{2}{\sqrt{5}}$

$\therefore \quad f'\left(\frac{1}{\sqrt{5}}\right)\left(-\frac{2}{\sqrt{5}}\right) = 3(5) - 2(5)(2).$

$= 15 - 20$

$f'\left(\frac{1}{\sqrt{5}}\right) = \frac{5}{2}\sqrt{5}.$

[27 Jun, 2022 (Shift II)]

68. (1)

Put $x^2 = u \Rightarrow 2x\,dx = du$

$\int \frac{du}{(u+1)(u+5)} = \frac{1}{4}\int\left(\frac{1}{u+1} - \frac{1}{u+5}\right)du$

[using partial fractions]

$f(x) = \frac{1}{4}\ln\left(\frac{x^2+1}{x^2+5}\right) + C$

$f(2) = \frac{1}{4}(\ln 5 - \ln 9) + C$

$\Rightarrow \quad C = 0$

$f(5) = \frac{1}{2}\ln\left(\frac{13}{15}\right)$

[25 Jan, 2023 (Shift-I)]

69. (4)

$I = \int\limits_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{7x + \frac{\pi}{2}}{2 - \cos 2x}dx \qquad \ldots.(i)$

$x \to -x$

$I = \int\limits_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{-7x + \frac{\pi}{2}}{2 - \cos 2x}dx \quad \left[\because \int\limits_{-a}^{a} f(x) = \int\limits_{-a}^{a} f(-x)\,dx\right] \ldots(ii)$

(i) + (ii)

$2I = \int\limits_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{2.\frac{\pi}{2}}{2 - \cos 2x}dx$

$$\Rightarrow \quad I = \frac{\pi}{2} \cdot 2\int_0^{\frac{\pi}{4}} \frac{dx}{2-\cos 2x}dx$$

$$\Rightarrow \quad I = \frac{\pi}{2} \cdot 2\int_0^{\frac{\pi}{4}} \frac{\left(1+\tan^2 x\right)dx}{2\left(1+\tan^2 x\right)-\left(1-\tan^2 x\right)}$$

$$\left[\because \cos 2x = \frac{1-\tan^2 x}{1+\tan^2 x}\right]$$

$$\Rightarrow \quad I = \pi\int_0^1 \frac{dt}{3t^2+1}$$

$$\Rightarrow \quad I = \frac{\pi}{\sqrt{3}}\tan^{-1}\sqrt{3}$$

$$\Rightarrow \quad I = \frac{\pi^2}{3\sqrt{3}}$$

[1 Feb, 2023 (Shift-II)]

70. (2)

$y^2 = 8x + 4y + 4$

$(y-2)^2 = 8(x+1)$

$Y^2 = 4aX$

Where, $a = 2$, $X = x + 1$, $Y = y - 2$

focus $\equiv (2, 0)$ and $x + 1 = 2$, $y - 2 = 0$

focus $(1, 2)$

$y - 2 = m(x - 1)$

Put $(3, 0)$ in the above line

$$\Rightarrow \quad m = -\frac{2}{3}$$

Length of focal chord = 26

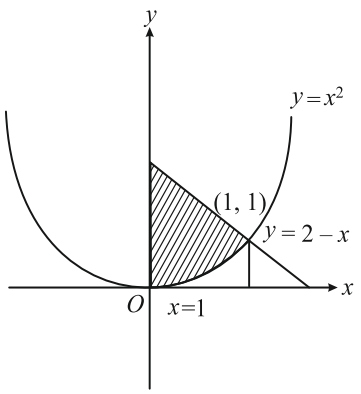
[1 Feb, 2023 (Shift-II)]

71. (13)

$$\left\{(x,y) \in R \times R, x \geq 0, x^2 \leq y \leq 2-x\right\}$$

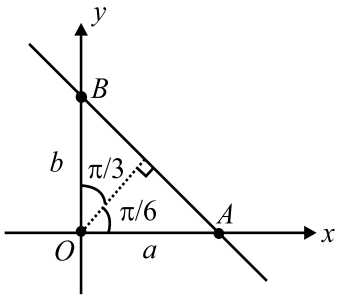
$$\therefore \quad \text{Required area} = \int_0^1 \left(2-x-x^2\right)dx$$

$$= \left(2x - \frac{x^2}{2} - \frac{1}{3}x^3\right)\Bigg|_0^1 = \frac{7}{6}$$



[25 July, 2021 (Shift-I)]

72. (3)



Equation of line intercept form: $\frac{x}{a} + \frac{y}{b} = 1$ ....(i)

Equation of line in normal form

$$x\cos\frac{\pi}{6} + y\sin\frac{\pi}{6} = p$$

$$\Rightarrow \quad \frac{\sqrt{3}x}{2} + \frac{y}{2} = p \Rightarrow \frac{x}{\frac{2p}{\sqrt{3}}} + \frac{y}{2p} = 1 \qquad \ldots(ii)$$

Comparing eqn (i) & (ii), we get $b = 2p, a = \frac{2p}{\sqrt{3}}$

Now area of $\Delta OAB = \frac{1}{2} \cdot ab = \frac{98}{3} \cdot \sqrt{3}$

$$\frac{1}{2} \cdot 2p \cdot \frac{2p}{\sqrt{3}} = \frac{98}{3} \cdot \sqrt{3}$$

$$\Rightarrow \quad p^2 = 49$$

$$\frac{b^2}{a^2} = 3$$

[30 Jan, 2023 (Shift-I)]

73. (65)

$$\because \quad x = \sin\left(2\tan^{-1}\alpha\right) = \frac{2\alpha}{1+\alpha^2} \qquad \ldots.(i)$$

and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right) = \sin\left(\sin^{-1}\frac{1}{\sqrt{5}}\right) = \frac{1}{\sqrt{5}}$

Now, $y^2 = 1 - x$

$$\frac{1}{5} = 1 - \frac{2\alpha}{1+\alpha^2} \Rightarrow 1+\alpha^2 = 5 + 5\alpha^2 - 10\alpha$$

$$\Rightarrow \quad 2\alpha^2 - 5\alpha + 2 = 0$$

$$\therefore \quad \alpha = 2, \frac{1}{2} \therefore \sum_{\alpha \in S} 8\alpha^3 = 8 \times 2^3 + 8 \times \frac{1}{2^3} = 65$$

[25 July, 2022 (Shift-II)]

74. (20)

Given, $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$

$$\vec{a} \times \left(\vec{c} - \vec{b}\right) = 0$$

$$\lambda\vec{a} = \vec{c} - \vec{b}$$

$$\Rightarrow \quad \vec{c} = \vec{b} + \lambda\vec{a} \text{ or } \vec{a} \cdot \vec{c} = \vec{a} \cdot \vec{b} + \lambda\left|\vec{a}\right|^2$$

$$\Rightarrow \quad -12 = (6\alpha + 75) + \lambda(261)$$

$$\Rightarrow \quad 2\alpha + 87\lambda = -29 \qquad \ldots.(i)$$

Given, $\vec{c}\cdot\left(\hat{i}-2\hat{j}+\hat{k}\right)=5$

$\because \quad \vec{c}=\vec{b}+\lambda\vec{a}$

$\vec{c}=\hat{i}\left(\alpha+6\lambda\right)+\hat{j}\left(11+9\lambda\right)+\hat{k}\left(-2+12\lambda\right)$

$\Rightarrow \quad (\alpha + 6\lambda) - 2\,(11 + 9\lambda) + (-2 + 12\lambda) = 5$

$\Rightarrow \quad \alpha - 24 = 5 \Rightarrow \alpha = 29$

From eqn. (i), $\lambda = -1$

$\vec{c}=\hat{i}\left(23\right)+\hat{j}\left(2\right)+\hat{k}\left(-14\right)$

$\vec{c}\cdot\left(2\hat{i}+\hat{j}+2\hat{k}\right)=20$

[8 April, 2023 (Shift-I)]

75. (386)

$\frac{1+3+5+a+b}{5}=5$

$a + b = 16 \quad \ldots.(1)$

$\sigma^2=\frac{\sum x_1^2}{5}-\left(\frac{\sum x}{5}\right)^2$

$8=\frac{1^2+3^2+5^2+a^2+b^2}{5}-25$

$a^2 + b^2 = 130 \quad \ldots.(2)$

by (1), (2)

$a = 7, b = 9$

or $a = 9, b = 7$

[01 Feb, 2023 (Shift-I)]

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